

Cubic Curves in PG(2, 4)

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The order of the group is 120960

The group has 0 orbits:

The overall number of objects is: 0

summary of the stabilizer orders:

Families 1, 2, 3, E, H:

e=0 iso1=-1 iso2=-1 iso3=-1 isoE=-1 isoH=-1

e=1 iso1=-1 iso2=-1 iso3=-1 isoE=-1 isoH=-1

e=2 iso1=-1 iso2=-1 iso3=-1 isoE=-1 isoH=-1

e=3 iso1=-1 iso2=-1 iso3=-1 isoE=-1 isoH=-1

c=1 d=1 isoG=-1

c=1 d=2 isoG=-1

c=1 d=3 isoG=-1

c=2 d=1 isoG=-1

c=2 d=2 isoG=-1

c=2 d=3 isoG=-1

c=3 d=1 isoG=-1

c=3 d=2 isoG=-1

c=3 d=3 isoG=-1

Distribution: ()

Curve	Ago	Pts	s. Pts	Infl	References
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with canonical forms

$$\begin{aligned}
 F1_e &= X^2Y + XY^2 + eZ^3 \\
 F2_e &= X^2Y + XY^2 + XYZ + eZ^3 \\
 F3_e &= XYZ + e(X + Y + Z)^3 \\
 E_d &= Z^2Y + X^3 - dY^3 \\
 H_e &= Z^2Y + X^3 + eXY^2 \\
 G_{c,d} &= Z^2Y + X^3 + cXY^2 + dY^3
 \end{aligned}$$

for $c, d, e \in \mathbb{F}_4$